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Fire Ecology

ORIGINAL RESEARCH

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Wildfire assessment using machine learning algorithms in different regions

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Abstract

Background Climate change and human activities are two main forces that affect the intensity, duration, and frequency of wildfires, which can lead to risks and hazards to the ecosystems. This study uses machine learning (ML) as an effective tool for predicting wildfires using historical data and influential variables. The performance of two machine learning algorithms, including logistic regression (LR) and random forest (RF), to construct wildfire susceptibility maps is evaluated in regions with different physical features (Okanogan region in the US and Jamésie region in Canada). The models' inputs are eleven physically related variables to output wildfire probabilities.

Results Results indicate that the most important variables in both areas are land cover, temperature, wind, elevation, precipitation, and normalized vegetation difference index. In addition, results reveal that both models have temporal and spatial generalization capability to predict annual wildfire probability at different times and locations. Generally, the RF outperforms the LR model in almost all cases. The outputs of the models provide wildfire susceptibility maps with different levels of severity (from very high to very low). Results highlight the areas that are more vulnerable to fire. The developed models and analysis are valuable for emergency planners and decision-makers in identifying critical regions and implementing preventive action for ecological conservation.

Keywords Wildfire, Fire regime, Machine learning, Random forest, Logistic regression, Fire Susceptibility map, Conservation ecology

Resumen

Antecedentes El cambio climático y las actividades humanas son dos de las fuerzas principales que afectan la intensidad, duración, y frecuencia de los incendios, lo que puede conducir a riesgos e incertidumbres en los ecosistemas. Este estudio usó la técnica de aprendizaje automático (*machine learning*) como una herramienta efectiva para predecir incendios de vegetación usando datos históricos y variables influyentes. La performance de dos algoritmos del aprendizaje automático, incluyendo regresiones logísticas (LR) y bosques al azar (*Random Forest,* RF), para construir mapas de susceptibilidad a los incendios, fue evaluado en regiones con diferentes características físicas (la región de Okanogan en los EEUU y la de Jamésie en Canadá). Los inputs del modelo fueron once variables físicamente relacionadas y cuyos resultados fueron las probabilidades de incendios.

Resultados Los resultados indican que las variables más importantes en esas dos áreas son la cobertura vegetal, la temperatura, el viento, la elevación, la precipitación, y el NDVI (Indice Normalizado de Vegetación). Adicionalmente, los resultados revelan que ambos modelos tienen la capacidad de generar espacial y temporalmente la predicción de la probabilidad anual de la ocurrencia de incendios en tiempos y ubicaciones diferentes. Generalmente, el RF excede

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al modelo LR casi todos los casos. Como resultado, el modelo provee de mapas de susceptibilidad con diferentes niveles de severidad (desde muy altos a muy bajos). Los resultados también resaltan las áreas que son más vulnerables al fuego. Los modelos desarrollados y el análisis son muy valiosos para los que planifican las emergencias y, para los decisores, para identificar regiones críticas e implementar acciones preventivas para la conservación ecológica.

Introduction

Forests cover one third of the planet and are vital natural resources to maintain ecological equilibrium in the environment (Gholamnia et al. 2020). The critical role of forest ecosystems lies in their natural ability to filter air and produce oxygen (Ljubomir et al. 2019). Nature conservation and human well-being are closely tied to natural forests and their associated ecosystems. Frequent wildfires, which ravage millions of hectares of global forested areas annually (Tehrany et al. 2021), pose a considerable threat to the ecosystem, species, and humans, resulting in a sustained disruption of the global ecological balance. Although wildfire is a common natural event in numerous ecosystems and not all wildfires result in disasters or catastrophes, they can pose risks, leading to damages to livelihoods, properties, and fire-sensitive ecosystems. They can also cause injuries and fatalities, particularly when they are amplified by factors like climate change, human activities, and invasive species (Tedim & Leone 2020). This critical issue has gained significant attention and concern from nations worldwide (Zhang et al. 2019).

Donato et al. (2023) compared the record-breaking large fires in recent years with historical active fire regimes. They showed that fires ignited by lightning and Native Americans played a crucial role in sustaining forest resilience over the years. In recent years, there has been a significant increase in the size and severity of wildfires in the US and Canada, partly due to climate change (Abatzoglou & Williams 2016; Flannigan et al. 2009). Indeed, climate change has widespread direct and indirect effects across the globe, leading to an increase in the frequency, intensity, and seasonal variation of extreme events (Moghim & Takallou 2023). Halofsky et al. (2020) integrated insights on the impacts of climate change and fire regimes on forests in the Pacific Northwest of the United States. Their findings indicated that large and intense fires in this region are linked to warm and dry weather, which is expected to become more common as the climate continues to warm. Projections based on historical data, present trends, and simulation models suggest that prolonged periods of warmth and dryness will lead to reduced fuel moisture and extended fire seasons in the future, likely resulting in more frequent and widespread fires compared to those in the twentieth century. Researchers can better recognize wildfires' underlying causes and dynamics in a particular region by evaluating climate conditions, vegetation type, topography, ignition sources, and human activities. Understanding a region's fire regime—characterized by frequent low-severity fires (low-intensity), rare high-severity fires (high-intensity), and mixed-severity fires (intermediate frequencies)—is crucial for studying wildfires as it provides precious insights into the historical patterns, frequency, and intensity of fires in that specific area (Agee 2003). To effectively study wildfires, it is necessary to comprehend the historical connections between fire activity and the factors that affect its occurrence, frequency, and intensity (Walsh et al. 2018).

Agee (2003) used the historical fire return intervals to evaluate the range of variability of the forest age structure in the central-eastern Cascade Range in Washington State, USA. Findings showed that early successional forest stages were more prevalent in high-elevation areas than the low-elevation ones. In addition, the proportion of old-growth and late successional forests ranged from 38 to 63% of the forested landscape. Walsh et al. (2018) investigated the multi-century (Holocene) fire regimes of low-elevation ponderosa pine-dominated forests in the Pacific Northwest, USA. They aimed to better understand the characteristics of past fire regimes, including fuel types and fire severity, as well as the likely ignition sources of those fires. Their findings suggest that while climate change over centennial to millennial timescales had minimal impact on the fire regimes at the study sites during the last 3800 years, the use of fire by Native Americans for various purposes, especially after approximately 1200 calendar years ago, had a significantly greater effect on the fire regimes. Dansereau and Bergeron (1993) evaluated the history of fire in the boreal forest of Western Quebec. Their results from different types of surficial materials indicate that environmental features have a greater influence on the delimitation of smaller fires in this region. A clear insight helps researchers and policymakers assess future fire behavior and possible hazards and thus develop effective strategies for fire monitoring and control systems (Gavin et al. 2007).

There are two main drivers of wildfires, including natural and anthropogenic forcing. Indeed, global warming, coupled with industrialization augmented by human activities, has led to a marked rise in the frequency and strength of wildfires worldwide (Gholamnia et al. 2020). These fires can adversely affect natural biological diversity, soil productivity, and carbon dioxide storage (Pham et al. 2020). The Okanogan region in the US and the Jamésie region in Canada experience a high frequency of wildfires, highlighting the need for a more in-depth examination, which is the focus of this paper. The Okanogan region, prone to fires due to a high frequency of lightning ignitions in June, July, August, and September, also experiences common human-caused fires throughout the region (Bloch et al. 2013a, b). Lefort et al. (2004) claimed that although both lightning and human-caused fires contribute significantly to the total area burned in the Jamesie region, humans are more responsible than lightning in increased fires. In addition, they stated that climatic factors such as summer precipitation and maximum temperatures play a primary role in fire occurrences and fire size, regardless of the ignition sources. Temperature rise can extend the duration of both fire and growing seasons, leading to increased evaporation, reduced soil moisture, and drying out of fuels, thus enhancing the chance of ignition and fire growth (Westerling 2016). In addition, a reduction in summer precipitation is projected to further increase fire spread, and the area burned (Holden et al. 2018).

Bloch et al. (2013a, b) used the historical fire regime in Okanogan County to show that many river basins experienced fires every 35 to 200 years in higher elevation forested regions, typically resulting in stand-replacing severity fires. In addition, they showed that some areas experienced fires over a longer period of over 200 years. However, most of the county, particularly the lower-elevation forests and shrub-steppe areas, experienced the same frequency of low to mixed-severity fires depending on the topographical features, amount, and timing of snowpack in those regions (Bloch et al. 2013a, b). The historical fire regime in the Jamésie region indicates that in Jamésie's boreal forest, the fire cycle is approximately 200 years, while in the eastern part, like Côte-Nord, it is significantly longer at about 500 years (Gauthier et al. 2011). However, the burned area and the number of large fires in the Jamésie fire regime zone have increased from 1959 to 2015 (Hanes et al. 2018). Table 1 summarizes the historical fire frequency in the study areas.

Fire hazard assessment can be characterized quantitatively or qualitatively as a marker of the probability of a given geographic region's combustion within a designated period. Fire hazard modeling is a complex tool that can be efficiently used for forest conservation and restoration (Tonini et al. 2020). Valid determination of the likelihood of fire occurrence and relevant susceptibility maps are essential in mitigating the detrimental impacts of such fires (Jaafari & Pourghasemi 2019). Different approaches, including expert knowledge, statistics, and machine learning (ML), are used to show regions susceptible to fire, commonly known as wildfire hotspots. Results based on expert knowledge strategy can be biased due to personal opinion (Tien Bui et al. 2016). Statistical methods, as another approach, use an extensive range of spatial data with diverse scales and resolutions, which can cover vast regions for wildfire modeling (Gigović et al. 2019). ML, as one of the data-driven models, tries to find patterns in the data and uses those patterns to make predictions about future data or other results of interest (Murphy, 2012). Chuvieco & Congalton (1989) and De Vliegher (1992) forecasted wildfire probability in Spain and Greece, respectively. These studies that are consistent with more recent works (e.g., Gigović et al. 2019 and Jaafari et al. 2019a) suggest that future fire incidents in each specific region are expected to occur under conditions similar to their earlier ones. Many researchers developed methods to construct a spatial model of fire susceptibility using remote sensing data and incorporating a diverse set of explanatory variables such as land, weather, and anthropogenic factors. For instance, Jain et al. (2020) comprehensively evaluated recent studies that used group-based classifiers like random forest (RF), boosted regression tree, artificial neural network (ANN), support vector machine (SVM), decision tree (DT), and dynamic Bayesian network for wildfire assessment. Their analysis shows that although wildfire modeling and prediction needs comprehensive approaches due to their complexity, the use of ML techniques for wildfire assessment has been increasing since their initial implementation in the 1990s.

Table 1 Fire frequency in the study areas

Okanogan		**Jamésie**	
Location	**Frequency**	**Location**	**Frequency**
Higher elevation forested regions	35 to 200 years with stand-replacing severity fires	Jamésie's boreal forest	200 years
Lower-elevation forests and shrub-steppe areas	35 to 200 years with low to mixed-severity fires	Eastern parts, like Côte–Nord	500 years

Several works compared different machine learning and statistical techniques in generating susceptibility maps of fire incidence. For instance, Adab (2017) showed that ANN outperforms the double logic regression model in examining fire hazards in the northeastern region of Iran. Guo et al. (2016) compared RF and logistic regression (LR) to prepare a fire susceptibility map in China. Their results demonstrated a superior outcome using RF. Ghorbanzadeh et al. (2019) compared the performance of the ANN, SVM, and RF to find the most effective model for spatial prediction of wildfire susceptibility. Their results showed that the RF model has the best performance. Gigović et al. (2019) used the Bayesian average approach to evaluate the performance of an ensemble learning method based on SVM and RF in constructing a fire susceptibility map. Their results showed a slightly better performance for the ensemble model than the single SVM or RF approach. Ngoc-Thach et al. (2018) compared the efficiency of SVM, RF, and multilayer perceptron neural networks (MLP-NN) to provide the fire susceptibility map in Vietnam. Their results indicated that although all methods performed similarly, MLP-NN showed the most substantial accuracy. Rodrigues and de la Riva (2014) compared four fire susceptibility mapping techniques, including RF, ERT, SVM, and LR. They showed that the RF could lead to the most accurate results with the minimum processing time. Tehrany et al. (2019) compared different methods, including RF, SVM, kernel logistic regression (KLR), and logit boost group-based decision tree (LBGb-DT), to produce a fire susceptibility map in Vietnam. Their results indicated that the LBGb-DT method has superior performance. Zhang et al. (2019) examined the performance of different ML techniques, such as convolutional neural networks (CNN), RF, SVM, ANN, and KLR, to construct a comprehensive regional fire susceptibility map in China. Their results showed that CNN can efficiently use spatial correlation to extract spatial features during the learning process, which leads to better performance.

A fire susceptibility map can indicate fire ignition and spread. Various machine-learning methods have been used to forecast fire growth using environmental and historical fire data (Huot et al. 2022). Markuzon and Kolitz (2009) evaluated various classifiers (RF, Bayesian networks, and K nearest neighbor) to predict fire growth one or 2 days after its detection. Their results showed that all models have almost the same performance with a high predictive accuracy using remote sensing images. The models showed an accuracy of over 75% in the correct classification of large fires, indicating that the classifiers effectively learned to identify the patterns associated with the large incidents. Similarly, Vakalis et al. (2004) employed an ANN with a fuzzy logic model based on various influencing factors, including terrain features, types and density of vegetation, and weather conditions, to assess the rate of fire spread in the mountainous area of Attica, Greece. Their findings indicate that both wind and topography significantly affect the spread rate as they create convective currents that interact with the fire front. To predict fire spread, Hodges and Lattimer (2019) used a CNN that forecasts spreading patterns based on environmental factors like topography and weather obtained from synthetic data. The results demonstrate that CNN can accurately predict the propagation of wildfires with less computational cost—three orders of magnitude less—than traditional modeling in heterogeneous spatial conditions. Radke et al. (2019) used a similar approach to anticipate fire spread using remote-sensing data obtained from geographic information systems (GIS), integrating both topography and weather conditions. Their results show that the model identifies patterns of wildfire spread by examining historical fire events. In addition, assessment of the model's outputs highlights the algorithm's ability to predict high-risk areas for fire spread up to 2 weeks ahead.

Although many studies evaluated the accuracy of the models for wildfire analysis, they did not address the physical features of the regions that can affect the accuracy of the results. In addition, no study compared the performance of different machine learning algorithms in locations with different physical features (e.g., topography and climate). Since the performance of different methods for wildfire assessment varies in different locations, this study compares two methods (LR and RF) to produce wildfire susceptibility maps over two regions with different climatic and topographical conditions, including Okanogan (USA) and Jamésie regions (Canada). In addition, this study evaluates the impacts of main parameters on the spatial distribution of the fire hazard over these regions. The most important parameters can clarify the underlying mechanisms responsible for the initiation of wildfires in such areas. Furthermore, this work examines the performance of the constructed models for annual wildfire probability prediction at different times and locations (temporal and spatial generalization ability). Analysis and results are valuable for detecting and monitoring critical regions to develop conservation plans for the ecosystem. The study area and datasets are explained in the "Study area" and Datasets" sections, respectively. The "Methodology" section explains the methodology. Results are explained in the "Results" section, followed by discussion and concluding remarks in the "Discussion and concluding remarks" section.

Study area

This work compares two models to evaluate their performance in producing wildfire susceptibility maps over two different regions. One of the case studies is Okanogan County in the upper part of Washington State in the United States, extending from latitude 47.95° to 49° north and longitude 118.82° to 120.88° west (Fig. 1). The Okanogan, which has an area of 13,664 square kilometers and a population of 42,000 in 2021, is the largest county in Washington. It has hot and arid summers with mainly clear skies and frigid, snow-laden winters marked by mostly cloudy conditions. The average minimum and maximum temperature varies between − 6 °C to 33 °C. The annual mean temperature is about 10 °C. The annual precipitation fluctuates between 358 and 800 mm. The Okanogan region displays modest seasonal variation in mean wind velocity, which ranges from 1.8 to 2.4 m s^{-1} over the year. The county encompasses a wide range of elevations, from the Columbia and Okanogan River valleys at around 200 m above sea level to the high peaks of the North Cascades at over 2750 m. Over the past few years, this region has experienced numerous wildfires. In July 2014, Washington State had a large wildfire event (Carlton Complex Fire) that extensively spread over the north-central area of Washington State, including Okanogan County. The Carlton Complex Fire has been the most massive wildfire recorded in the history of Washington State (Halofsky et al. 2020). The wildfire, initiated by four distinct lightning strikes, affected over 1011 square kilometers of public and private property, resulting in a total estimated damage cost of 30 million dollars (Furman 2016).

Another case study is in the western terrain of Jamésie in west Quebec and spans across latitudes of 51.31° to 53.8° north and longitudes of 78° to 79.15° west, located along the eastern coast of James Bay (Fig. 1). The Jamésie region contains a vast area of 283,955 square kilometers and the population of about 13,000 in 2021. This region includes a small variability of elevations, from 0 m to high peaks of around 210 m. Summers have cool temperatures and frequently overcast skies, whereas the winter season is marked by frigid cold spells, snowfall, stormy weather, and a foggy atmosphere. There is a notable range of annual cycle of temperature varying from − 27 °C to 20 °C. The annual average temperature is approximately − 2 °C, the mean annual precipitation is about 700 mm,

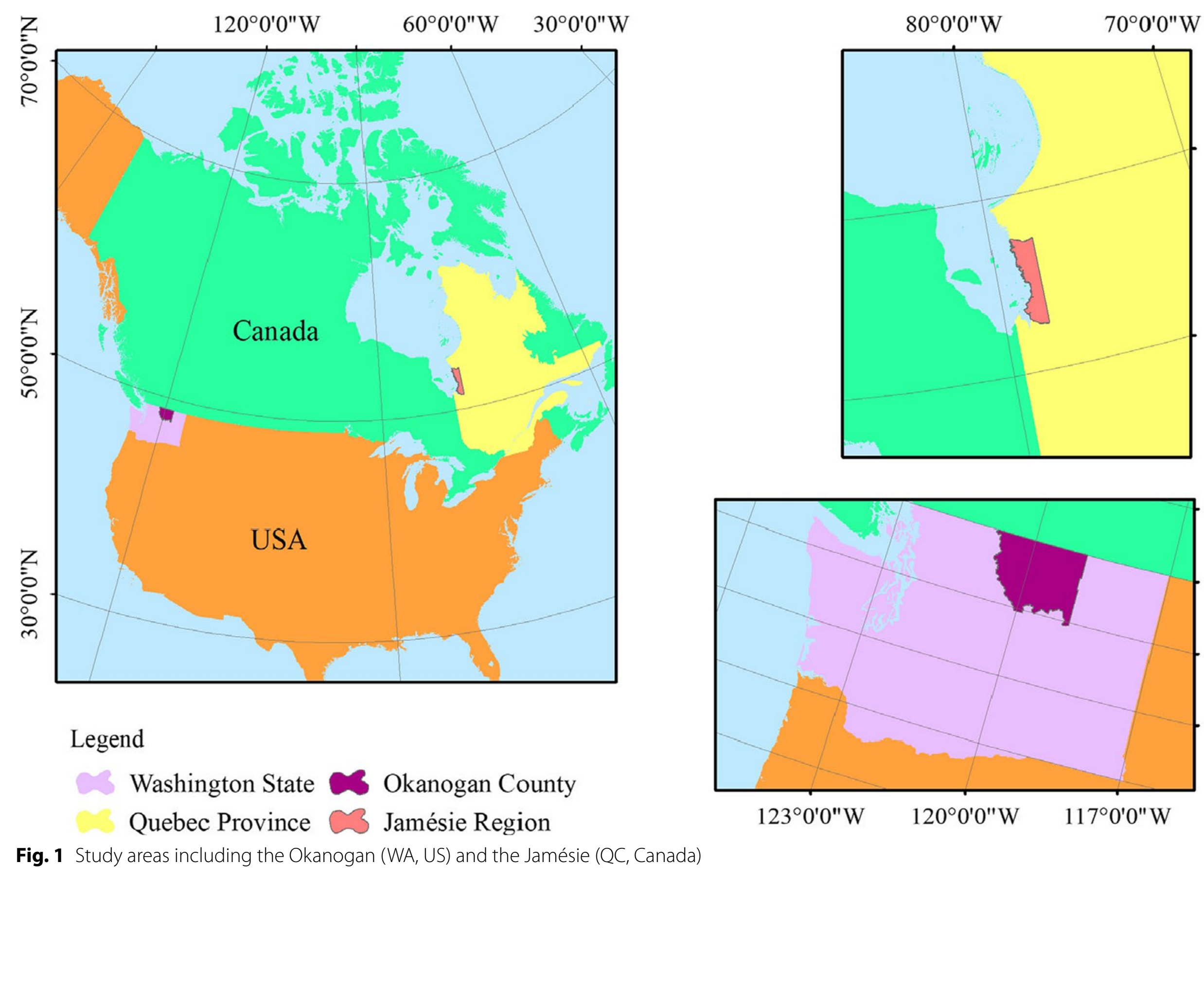

Fig. 1 Study areas including the Okanogan (WA, US) and the Jamésie (QC, Canada)

and the average wind velocity is about 14 m s^{-1}. Quebec's central and western regions have been prone to numerous wildfires recently. In July 2013, smoke emissions from a wildfire near James Bay in Quebec affected the air quality. The wildfire led to the combustion and destruction of about 2500 square kilometers (Camp et al. 1997). Due to these domains' different climates and physical features, the Okanogan and Jamésie are selected to evaluate the impacts of different parameters on wildfire.

Datasets

Datasets are the main requirement for the statistical and data-driven methods. Collecting valid data and developing a robust database can significantly affect the analysis (Pourtaghi et al. 2016). Different factors including topographic features, meteorological conditions, vegetation cover, and anthropogenic activities influence wildfires (Tien Bui et al. 2018), which can be prioritized based on locations (Gholamnia et al. 2020). This study uses a data-driven approach to synthesize 11 important parameters, including elevation, aspect, curvature, slope, temperature, precipitation, wind, land use land cover (LULC), normalized difference vegetation index (NDVI), and distance to roads and distance to rivers for each study area (Figs. 2 and 3) to predict fire susceptibility.

To establish a valid model using ML, an adequate and reliable classified dataset is essential. The historical wildfire inventory, which determines past burned areas on Earth, can be obtained from the Moderate Resolution Imaging Spectroradiometer (MODIS) satellite. MODIS is an operational device on the Terra and Aqua spacecraft. The apparatus is equipped with the electromagnetic spectral bands of both visible and thermal infrared (Junpen et al. 2013), providing a reliable archive of past fire occurrences with more than 90% fire detection accuracy (Tanpipat et al. 2009). The dataset used for this work is MOD14A1 Version 6, mainly derived from MODIS 4- and 11-μm radiance with 1-km spatial resolution, global coverage, and daily temporal resolution (data are publicly available at https://www.earthdata.nasa.gov/firms). The fire detection method relies on both absolute detection of a fire (when the fire is strong) and relative detection determined by surface temperature variability and sunlight reflection. The dataset contains information on fire occurrences (day/night), fire location, and detection confidence (Giglio & Justice 2015). Since a large amount of valid data can make a robust model, this study also uses GPS data from the Forest Protection Organization (data are publicly available at https://data-nifc.opendata.arcgis.com for the Okanogan and https://open.canada.ca for the Jamésie). The dataset consists of a compilation of wildfire data containing the specific locations of the fires (point data) and the boundaries of the fire areas (polygon data). Both datasets are integrated to generate highly accurate historical wildfire maps to determine areas with or without fire for each study area.

One of the factors that affects wildfires is topographic features (Gibson et al. 2020), which include elevation, slope, aspect, and curvature of the region (Pourghasemi et al. 2020; Shabani et al. 2020). The topographical characteristics dataset is obtained from the United States Geological Survey database (data are publicly available at https://earthexplorer.usgs.gov). The Global Digital Elevation Model (GDEM) Version 3, produced by the Terra Advanced Spaceborne Thermal Emission and Reflection Radiometer (ASTER), provides a digital elevation model (DEM) of the Earth lands. This topographical dataset has a spatial resolution of 1 arc second (about 30 m on the equator), multi-year temporal resolution, and global coverage.

Land cover is another physical feature of the region that is important in vulnerability and the flammability of the land. The LULC data include different types of land, such as forests, agriculture, impermeable land, snow, ice, wetlands, and shrubland (Pham et al. 2020). The Multi-Resolution Land Features Consortium (MRLC) consists of various federal agencies collaborating in developing and producing precise and appropriate land cover information through the National Land Cover Database (NLCD). The product is prepared by the Landsat satellite imagery with supplementary datasets. The NLCD provides comprehensive data related to the land cover in 16 classes and its variation across the USA with a spatial resolution of 30 m (Dewitz, 2021; Homer et al. 2020; Jin et al. 2019; Wickham et al. 2021; Yang et al. 2018). For the periods of 2001–2009 and 2010–2018 in the Okanogan, this study uses land cover data from 2006 and 2015, respectively (data are publicly available at https://mrlc.gov). Similar to the NLCD of the USA, the Canadian Center for Remote Sensing (CCRS) generated national and continental land cover products from different satellites (Pasos 2010). For the Jamésie region, this study uses the MODIS 2005 and Landsat 2015 land cover product for the 2001–2009 and 2010–2018 periods, respectively (data are publicly available at http://www.cec.org). The land cover dataset from MODIS 2005 version 3 has 19 classes with a spatial resolution of 250 m. The land cover dataset from Landsat 2015 version 3 has 19 classes with a spatial resolution of 30 m. One of the indicators of the LULC is NDVI, which can be expressed using the data from the Landsat 7 Enhanced Thematic Mapper Plus (ETM+) satellite (data are publicly available at https://earthexplorer.usgs.gov). Landsat 7 ETM+C2 L1 images comprise eight spectral bands, with a spatial resolution of 30 m for bands 1 to 7. Band 8, the panchromatic band, has a higher resolution of 15 m. The scene size is approximately 170 km in the north–south

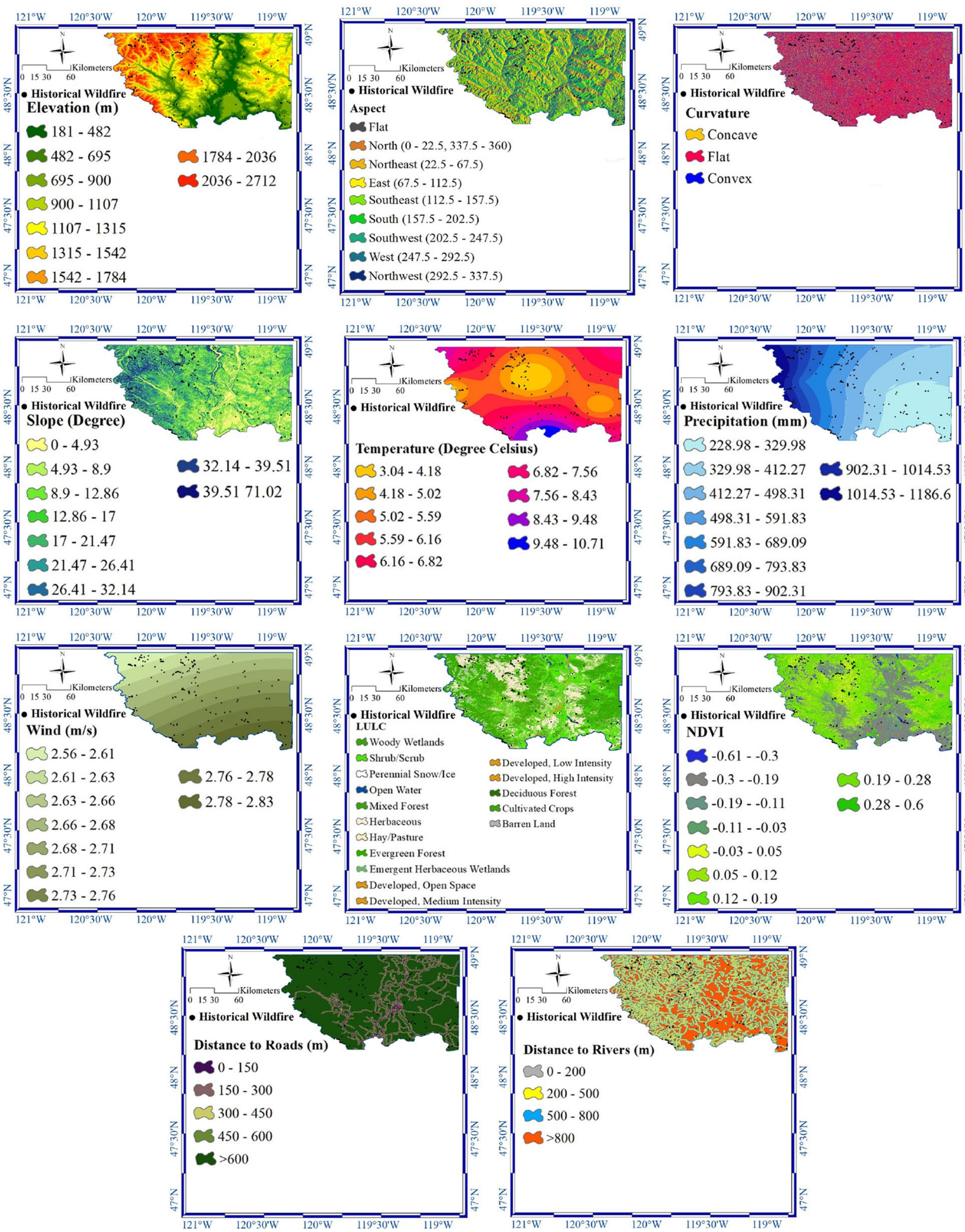

Fig. 2 Important parameters for wildfires in the Okanogan region

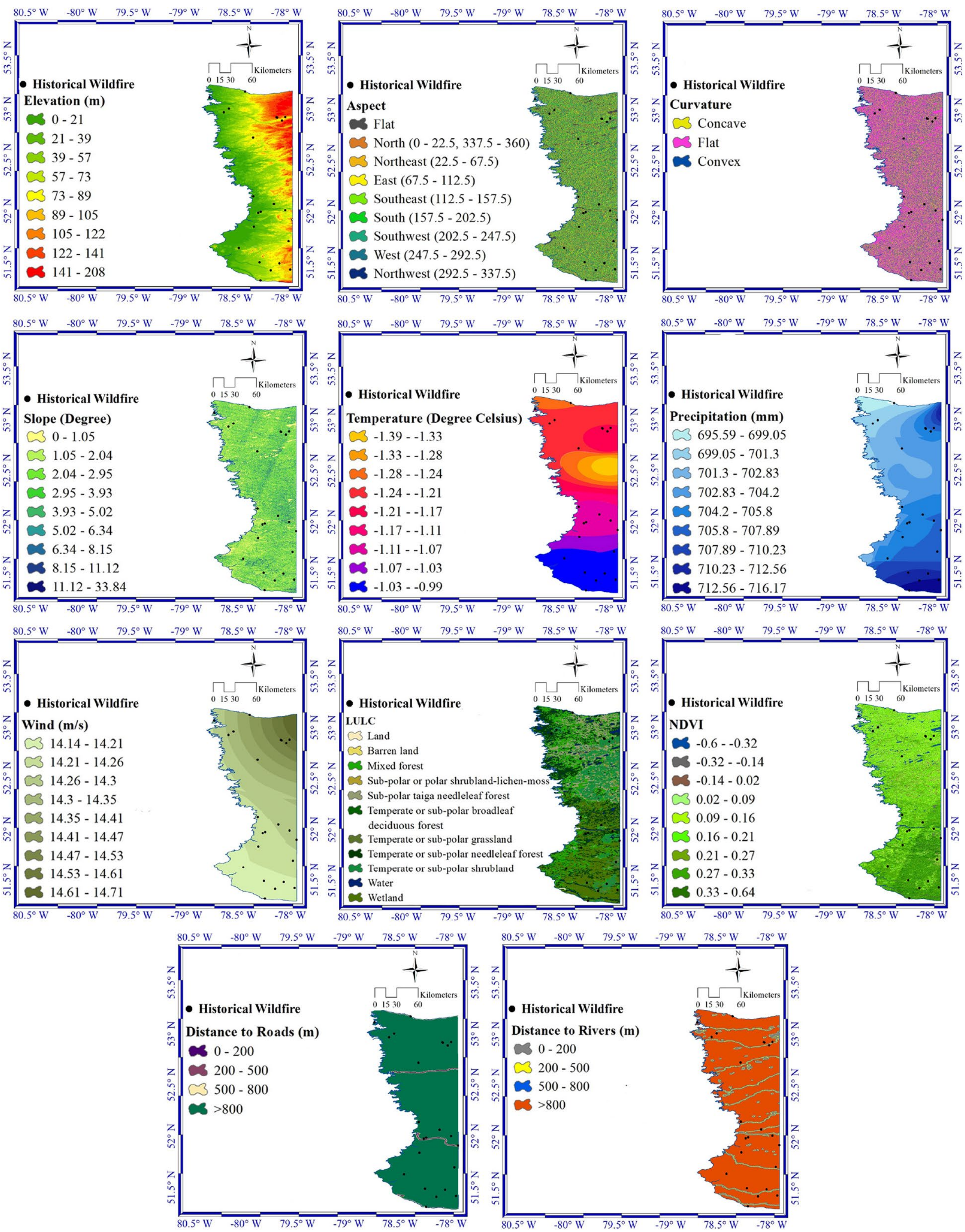

Fig. 3 Important parameters for wildfires in the Jamésie region

direction and 183 km in the east–west direction, with global coverage and 16 days temporal resolution. Since late fall and winter satellite images are mainly covered with ice and snow, and a significant reflection of the near-infrared light by the snow and ice can affect the accurate assessment of the vegetation density, as NDVI relies on near-infrared and red light reflectance, we excluded those satellite images in NDVI calculation. Thus, an average of the NDVI values from available Landsat 7 ETM+images in all months (excluding late fall and winter) of the years during the periods of 2001–2009 and 2010–2018 is used to calculate annual mean NDVI.

Another factor that affects wildfires is climate conditions defined by hydrometeorological variables like temperature, humidity, precipitation, and circulation changes (Adab et al. 2018). The hydrometeorological variables are obtained from meteorological stations (data are publicly available at https://ncei.noaa.gov and https://climate.weather.gc.ca for the Okanogan and Jamésie, respectively). Daily and monthly climatological data from different weather stations distributed in the study areas are collected and averaged to find annual mean data. An initial data screening found a few stations with invalid data, which were excluded from the process. This study employs the inverse distance weighted (IDW) method (Shepard 1968) to interpolate annual average climatic data to the pixels that lack data.

Anthropogenic factors can also trigger wildfires. Public access to the forested domains can increase wildfire vulnerability. Related data to the roads and rivers are extracted from the respective transportation and meteorological databases to create the river and road maps (data are publicly available at https://transportation.gov and https://weather.gov for the Okanogan and https://open.canada.ca for the Jamésie). The datasets are maps developed by the Bureau of Transportation Statistics for transportation networks to show the roads and the National Weather Service for streams to show the river lines throughout the study areas. Although rivers play an essential role in enhancing the greenness of the plants and surrounding land covers, reflected in the LULC and NDVI variables (Nagler et al. 2020), they are popular recreational spots for people, leading to increased human activities near these water bodies (Rajabi et al. 2013). To analyze the impacts of human recreational activities (e.g., camping site distances from roads and rivers) on the initiation of wildfires and finding more vulnerable areas, polygons are generated around the rivers and road lines.

Table 2 summarizes all datasets with the sources used for both study areas in this work.

Methodology

This study uses data-driven models to assess the performance of the logistic regression (LR) and random forest (RF) models as two widely used ML algorithms (Jain et al. 2020) to predict future wildfires using effective features. The basic concept of the models is to construct the inputs-output relationship that can properly predict wildfires considering different climates and physical features. The performance of the developed models depends on the predictors (inputs) that can characterize the events. Correlated variables and multicollinearity can substantially lower the model's accuracy (Jaafari et al. 2019b). In multicollinearity, one predictor variable can predict outcomes as accurately as others in a multiple-regression analysis. To assess the multivariate correlation between effective variables (predictors), this study uses variance inflation factors (*VIF*) and tolerance (*Tol*) as (Pourghasemi et al. 2020)

$$TOL_i = 1 - {R_i}^2 \tag{1}$$

$$VIF_i = \frac{1}{1 - {R_i}^2} \tag{2}$$

where ${R_i}^2$ is the regression's coefficient of determination for the explanatory variable i (Pourghasemi et al. 2020). A VIF equal to or larger than five or a tolerance value of

Table 2 Datasets used in this study

Variable	Source	Access source
Elevation, slope, aspect, curvature	USGS GDEM Version 3	Okanogan and Jamésie: www.earthexplorer.usgs.gov
LULC	MODIS and Landsat Images	Okanogan: www.mrlc.gov Jamésie: www.cec.org
NDVI	Landsat 7 ETM+C2 L1 images	Okanogan and Jamésie: www.earthexplorer.usgs.gov
T, P, wind	Meteorological Organizations	Okanogan: www.ncei.noaa.gov Jamésie: www.climate.weather.gc.ca
Distance to roads	Transportation Organizations	Okanogan: www.transportation.gov Jamésie: www.open.canada.ca
Distance to rivers	Meteorological Organizations	Okanogan: www.weather.gov Jamésie: www.open.canada.ca

less than 0.2 indicates multiple correlations (Hong 2019). Variables that meet these criteria should be excluded from inputs, which can help find proper predictors (with lower interdependency).

One of the commonly used ML algorithms is LR, which is a simple, fast, and popular classification method (Jain et al. 2020). LR applies linear regression in the logit scale to identify the most robust linear combination of variables most likely to detect the observed outcome (Stoltzfus 2011). It measures the individual impact of each variable to evaluate the effect of a set of independent variables in a binary outcome.

The LR can determine the relationship between binary response (0 for no fire and 1 for fire) and possible effective variables that can drive the model (fire occurrence and related probabilities) (Bisquert et al. 2012; Guo et al. 2016). We can evaluate the probability (*P*) of a fire occurrence ($Y=1$) using logistic coefficients and an S-shaped curve with values between zero and one as

$$P(Y=1)=\frac{1}{1+e^{-\alpha}} \tag{3}$$

Logistic regression includes an equation to fit the linear combination of the available data represented by α as

$$\alpha=\beta_0+\sum_{j=1}^{p}\beta_jX_j=\beta_0+\beta_1X_1+\beta_2X_2+\cdots+\beta_pX_p \tag{4}$$

where β_0 and β_i are the model intercept and coefficients $(i=1,2,3,\ldots)$ and X_i represents the effective variables (predictors). An increase in the model coefficient can lead to a higher likelihood of wildfires (Tehrany et al. 2014).

Another ML algorithm is RF, which is more elaborate. Breiman's random forest model is one of the ML techniques, encompassing a collection of simple trees that function cohesively (Breiman 2001; Yu 2017). The RF can construct a model for a particular predictant depending on independent predictors (Gigović et al. 2019). Two main essential parameters in the model include the number of trees and the number of variables for each node, the latter of which can be determined by the square root of the number of effective variables. The parameters are selected based on the optimal performance of the model and minimum errors. This study uses 400 and 300 trees for the Okanogan and Jamésie regions, respectively, and four sampling variables at each node for both areas. The number of trees is selected based on minimizing the models' error. Generally, increasing the number of trees can improve results to some extent, while after a specific number of trees, this improvement starts to decrease with adding more trees. In addition, adding more trees increases computational time, cost, and memory usage. The model's performance has a minimum change after 400 and 300 trees for two case studies, so these numbers are selected as an optimal value with minimum computational time and memory usage. Although there are many hyperparameters in the RF, this study focuses on the number of trees and the number of variables for each node as the two main parameters for the models' accuracy. For instance, the "Improve" parameter (default = 0.05) dictates the minimum improvement in the "out-of-bag" (OOB) error required for the search to proceed, and the "replace" parameter determines the sampling of cases with (default = True) or without replacement, and the "doBest" parameter, which specifies whether the model uses an optimal number of variables for each node or not (default = FALSE). The developed model uses arbitrary binary trees for the underlying framework and conducts bootstrapping techniques to determine a subset of observations for analysis. The training data are selected randomly, and the remaining data are identified as "OOB" data (Catani et al. 2013). The RF algorithm calculates the ultimate class membership through a vote count procedure across all decision trees and selects the class with the highest number of votes (Micheletti et al. 2014).

The RF examines the impact of different variables on the model's performance using the "mean decrease accuracy" or the "mean decrease Gini" method. The "mean decrease accuracy" is better than the "mean decrease Gini," particularly with the environmental variables (Arabameri et al. 2018). Therefore, this work uses the mean decrease accuracy to find the variables that strongly influence the results. The RF analyzes the effect of a specific variable on the model's skill by manipulating a small sample size through permutation of the variables' values and assessing the "mean decrease accuracy" (Guo et al. 2016). The variables with the highest "mean decrease accuracy" are selected as the most important.

The models' inputs are variables that can drive or affect the wildfires, including elevation, aspect, curvature, slope, temperature, precipitation, wind, LULC, NDVI, distance to roads, and distance to rivers. The output of the models is the annual probability of the fire occurrence. A proper set of predictors can construct a robust and efficient inputs-output model. Indeed, variables related to the physical mechanism of the event can lead to a valid and robust model. One of the predictors can be topographic features that affect hydrometeorological variables such as absorbed solar radiation, temperature variations, evapotranspiration, and wind. Indeed, topographic factors can have associative and indirect effects on the incidence of wildfires (Camp et al. 1997). For instance, the slope of the region influences the fire dispersion rate, and the maximum spread rate occurs in steep inclines (Ngoc Thach et al. 2018). In addition, the aspect can affect the moisture content of the fuel (Nyman et al. 2015), and the curvature impacts the behavior of the wildfires such that lower curvature leads to a greater fire incidence (Hilton et al. 2016). Another important feature that can affect wildfire incidence is LULC, which can also be expressed by NDVI. This index can indicate water and nutrient variability, plant diseases, and other stress factors, which are important in the vulnerability of vegetation to fire incidence (Bajocco et al. 2015). NDVI, which varies between negative and positive one, can quantify the greenness, density, and health of vegetation and can be calculated as

$$NDVI = \frac{NIR - RED}{NIR + RED} \tag{5}$$

where RED (red wavelength) and NIR (near-infrared wavelength) are bands 4 and 3 of Landsat 7 ETM+, respectively (Masek et al. 2006). The NDVI values closer to 1 correspond to healthier and dense plants, and values closer to -1 refer to non-vegetated surfaces (Jodhani et al. 2024).

In addition to the physical features of the regions, hydrometeorological variables that define climatic conditions affect wildfires. For instance, an increase in average rainfall with a decrease in average temperatures can cause an increased yield of vegetation and fuel generation, which provides valuable fuel resources for subsequent wildfires (Nunes et al. 2016). The wind plays a vital role in providing a significant amount of oxygen and thus accelerates the fire spread in vegetative fuels, facilitating the transmission and spread of wildfires. Therefore, the fire is susceptible to the surrounding wind, which drives the direction of the fire spread (Ljubomir et al. 2019; Pourghasemi 2016). The other driver of the wildfires is anthropogenic forcing. For instance, easy access to the forest can increase the incidence of wildfires, and the hazard of fire is higher near roads (Pourtaghi et al. 2016). In addition, rivers can contribute to increased wildfire incidents by serving as recreational areas (Rajabi et al. 2013); on the other hand, they can help mitigate the severity of fires by enhancing moisture levels in nearby vegetation (Kalantar et al. 2020).

All inputs are converted to different classes from one to 16, depending on the variable. The classes are identified through pixel-based histograms of the maps using the Natural Breaks technique, in which the features are sorted into groups based on their differences (North 2009). Natural Breaks classification aims to minimize variability within groups and maximize variability between groups, leading to internally consistent and externally diverse classes. This optimization process can make clear and separate classes that accurately reflect the underlying data distribution, and thus it is efficient in visual interpretation. In addition, Natural Breaks classification is responsive to the intrinsic structure of the data distribution, which can adapt to various data distributions and determine optimal class boundaries based on specific data characteristics. This adaptability enables Natural Breaks to capture intricate patterns and variations in the data more effectively than fixed interval or quantile-based classifications (Chen et al. 2013).

The developed models can evaluate the impact of predictors on wildfire occurrences and provide wildfire susceptibility maps in the two study regions in the USA and Canada. To investigate the generalization ability of the developed models, we use different sets of training data. To construct a robust model and find the most important factors on the performance of the models, 2001–2009 datasets are divided randomly into two groups; the first group is for training of the model (training data) encompassing 70% of the occurrences, and the rest of the data (30%) is for the model validation (validation data). To examine the temporal generalization capability of the models, the LR and RF are trained based on the 2001–2009 datasets and validated based on the 2010–2018 datasets. To evaluate the spatial generalization ability of the models, the developed models are trained based on the datasets from the Okanogan region (2001–2009) and validated based on the datasets from the Jamésie region (2001–2009) and vice versa.

The developed models generate the annual probability of wildfires based on effective variables (predictors) at each specific pixel in the two case studies. The probabilities provide wildfire maps indicating different classes of fire susceptibility, including very low, low, medium, high, and very high. The natural break classification is used to classify the regions based on the degree of sensitivity (Moayedi et al. 2020).

To evaluate the performance of the constructed models and prediction accuracy, this study uses the receiver operating characteristic (ROC) curve approach (Hanley & McNeil 1982). The approach assesses the model's effectiveness using success and prediction rate curves and ensures that the modeling procedure is robust through a comprehensive and meticulous analysis (Nami et al. 2018). The ROC method represents sensitivity rates or true positive rates on the y-axis versus 100-specificity or false positive rates on the x-axis (Hanley & McNeil 1982). This method can provide a quantitative measure that is able to properly distinguish between positive and negative outcomes and thus estimates model performance (Kalantar et al. 2020). The outputs of the models can provide wildfire maps based on the probabilities of the wildfire occurrences. To evaluate wildfire maps, this study uses the area under the receiver operating curve (AUC), which is widely used for examining the efficacy and capability of the classification models (Gigović et al. 2019; Jaafari et al., 2019a; Malik et al. 2021; Al-Juaidi et al. 2018). AUC values below 0.6 signify an inadequate performance, between 0.6 and 0.7 indicate a moderate level of performance, from 0.7 to 0.8 highlights a good performance, and values of 0.8 to 0.9 and greater than 0.9 signify a high and an outstanding level of performance, respectively (Hong 2019).

Results

Multicollinearity analysis (VIF method) between effective factors can find a set of predictors with the most remarkable impact on the likelihood of a fire event. The VIF method is used to assess the multivariate correlation between influential variables. Table 3 shows the variance inflation factor and tolerance coefficient for the two study regions. The VIF value of 1 indicates no correlation between the predictors. A higher VIF can refer to a larger correlation between one variable and the others, and a VIF larger than 5 indicates multicollinearity within the variables (Kalantar et al. 2019).

Results show that wind/precipitation and temperature/precipitation have the maximum VIF values in the Okanogan and Jamésie regions, respectively. Since, for all variables, the specified threshold is satisfied (the VIF smaller than five or a tolerance value equal/larger than 0.2), all variables are used as inputs in the developed models. To evaluate the relative significance of variables for wildfire occurrence, the RF model uses the "mean decrease accuracy" (MDA) method. Table 4 summarizes the results of MDA for both regions in the RF model.

There is a direct relationship between the importance level and the extent to which a variable's accuracy is reduced (Williams 2011). Table 4 shows that the most effective variables for wildfire in the Okanogan region are wind, precipitation, temperature, elevation, LULC, and NDVI, while precipitation, temperature, wind, LULC, elevation, and NDVI are the most important ones in the Jamésie region. Results indicate that the variables related to climate, vegetation, and elevation have a larger impact on the performance of wildfire prediction than anthropogenic and other topographic variables.

The outputs of the models that can generate the annual probability maps reflect the area's susceptibility to potential fire occurrences. In other words, the maps display the annual probability for each pixel as the degree of vulnerability to potential fires based on conditional factors. The susceptibility maps from each model that represent the distinctive levels of probabilities of wildfire occurrence (very high, high, moderate, low, and very low) across the study areas are shown in Fig. 4.

The performance of the models for the probability maps can be evaluated using validation data (30% of the unseen data in training). To examine the validity of the maps from two models, the receiver operating curve (AUC) is shown in Fig. 5. Results show that the performance rates of the LR in the Okanogan and Jamésie regions are 75.3% and 87.5%, respectively. Similarly, the performance rates of the RF in the Okanogan and the Jamésie regions are 98.2% and 99.1%, respectively. Results confirm that although wildfire prediction is complex and uncertain due to the interaction of different factors and variables, the developed models can predict wildfires properly.

Table 3 Multicollinearity assessment

Variable	Study area			
	Okanogan		Jamésie	
	VIF	Tolerance	VIF	Tolerance
Wind	4.73	0.21	3.58	0.28
Temperature	1.53	0.66	4.91	0.2
Slope	1.34	0.74	1.02	0.98
Precipitation	4.42	0.23	4.7	0.21
NDVI	1.31	0.76	1.33	0.75
LULC	1.15	0.87	1.15	0.87
Elevation	2.65	0.38	2.66	0.38
Distance to roads	1.24	0.81	1	1
Distance to rivers	1.12	0.89	1.11	0.9
Curvature	1	1	1.01	0.99
Aspect	1.01	0.99	1.01	0.99

Table 4 Relative importance of variables in the RF model

Variable	MDA	
	Okanogan	Jamésie
Wind	0.24	0.16
Temperature	0.21	0.17
Slope	0.02	0
Precipitation	0.23	0.21
NDVI	0.06	0.02
LULC	0.07	0.13
Elevation	0.15	0.09
Distance to roads	0.03	0
Distance to rivers	0.03	0.01
Curvature	0	0
Aspect	0.02	0

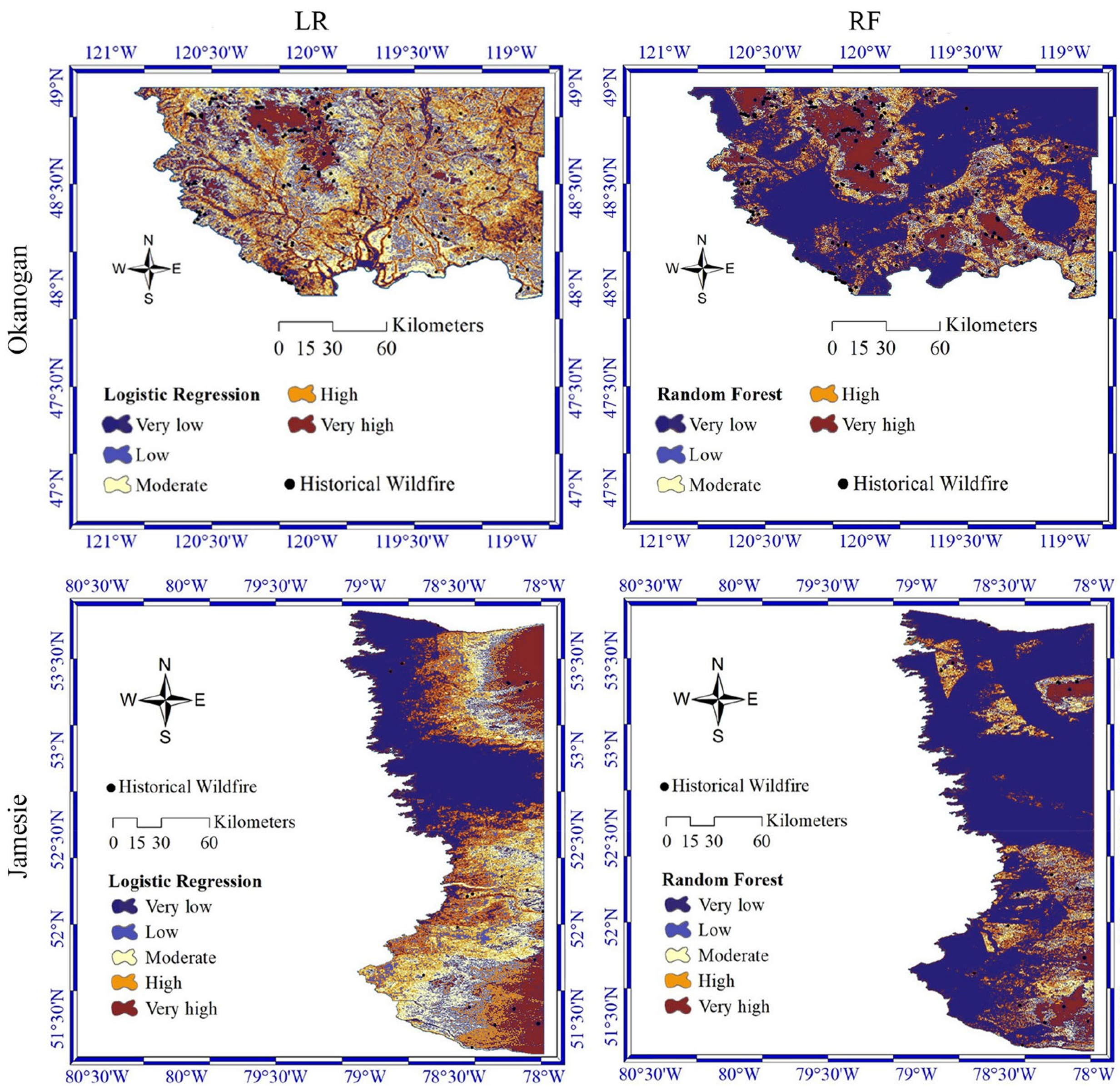

Fig. 4 Susceptibility maps from LR (left column) and RF (right column) in Okanogan (first row) and Jamésie region (second row)

To better assess the skill of the LR and RF in wildfire prediction, the models are trained using data from 2001 to 2009 and then validated with 2010–2018 datasets in both case studies. Results show that the performance rates of the LR for the validation period are 54.6% and 69.8% in the Okanogan and the Jamésie regions, respectively. Similarly, the performance rates of the RF are 55.5% and 75.5% in the Okanogan and the Jamésie regions, respectively. Results show that although LR and RF models were trained and validated by different times of datasets (temporal generalization capability), they have an accepted performance, particularly in the Jamésie region.

In addition to the temporal generalization ability of the models, this study examines the spatial generalization capability of the LR and RF (by training in one region and validating in another region). Results show that the success rates of the LR and RF models for predicting wildfires in the Jamésie region using training sets in the Okanogan region are 55.2% and 66.3%, respectively. Similarly, the accuracy levels of the LR and RF trained in the Jamésie region to predict wildfires in the Okanogan are 54.7% and 52.5%, respectively. Results

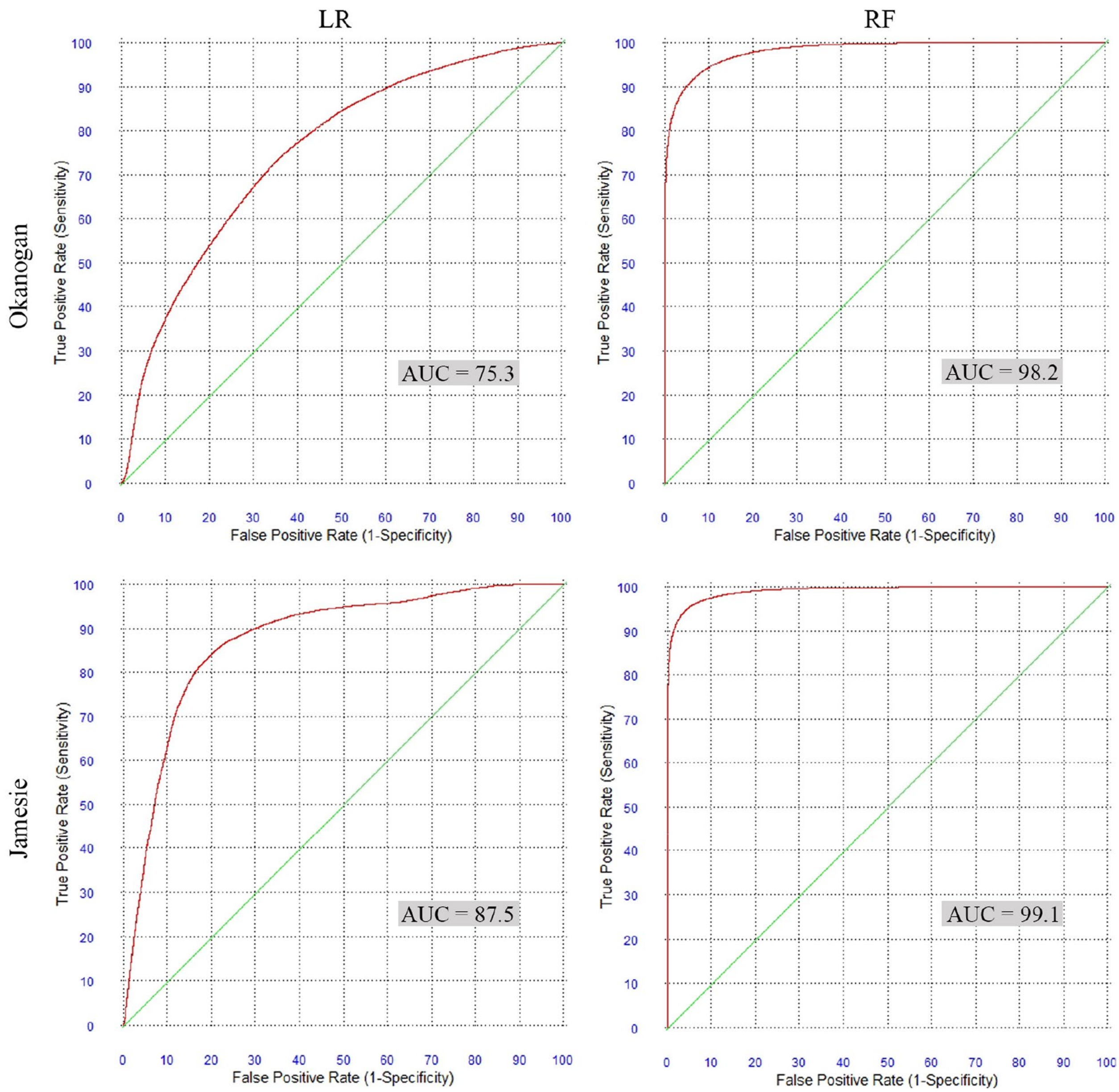

Fig. 5 Receiver operating characteristics (ROC) curves for LR (left column) and RF (right column) in Okanogan (first row) and Jamésie region (second row)

show that although models are trained with different climates and physical features, they can predict wildfire occurrences in different regions. Note that it is important to evaluate the performance of the constructed models differently. For instance, we have a larger AUC for the models that are trained and validated in the same locations, such as training the models in Okanagon and validating them in Okanagon. On the other hand, to evaluate the models' spatial generalization capability, for instance, training the models in the Okanagan and validating them in the Jamésie region, a smaller AUC can be acceptable and satisfactory when there is limited or no data available for the tested regions. In other words, the models can be trained in a location with available data and then applied to other areas with limited datasets. In this case, at least an estimation of wildfire occurrences can be examined. Table 5 summarizes corresponding AUC values for the performance of the models using different training sets. Results show that although the two models can perform well using different training sets, RF works better in most training-validation scenarios, especially in the Jamésie region.

Table 5 Performance of the models (LR and RF) in Okanogan and Jamésie regions

Training	Validation	Okanogan		Jamésie	
		LR	RF	LR	RF
70% of 2001–2009 datasets	30% of 2001–2009 datasets	75.3	98.2	87.4	99.1
2001–2009 datasets	2010–2018 datasets	54.6	55.5	69.8	75.5
2001–2009 datasets of the Okanogan	2001–2009 datasets of the Jamésie	-	-	55.2	66.3
2001–2009 datasets of the Jamésie	2001–2009 datasets of the Okanogan	54.7	52.5	-	-

Discussion and concluding remarks

Wildfires significantly threaten ecosystems, human lives, and infrastructure worldwide. As the frequency and intensity of wildfires continue to rise, there is an urgent need for effective prediction and prevention strategies. Emergency preparedness and prevention strategies rely heavily on wildfire maps, which depict possible locations of fires and their potential impacts on ecosystems and infrastructures, making them an essential tool for predicting wildfire outbreaks (Haas et al. 2013). This study explores the importance of machine learning (LR and RF) as a proper tool for wildfire prediction and finding effective parameters in wildfires in regions with different climate and physical features. To the authors' knowledge, this is the first study to quantitatively evaluate the performance of different machine learning algorithms in regions with different physical (e.g., topographic and climatic) features.

To develop a robust inputs-output model in LR and RF, a comprehensive set of topographic, climatic, and anthropogenic factors and vegetation features are combined with the historical burned areas to train and validate the models in Okanogan and Jamésie regions. The models' predictant (target or output) is the annual probability of fire occurrence based on the predictors (inputs), including elevation, aspect, curvature, slope, temperature, precipitation, wind, LULC, NDVI, distance to roads, and distance to rivers.

To construct valid and proper models, the multicollinearity test (VIF and tolerance) confirmed no remarkable linear relationship between explanatory data. To evaluate the performance of the models, the impact of the effective factors, and their importance in the performance, datasets in 2001–2009 are divided into two groups, including training (70%) and validation (30%) datasets. Results indicated that both models perform well in wildfire prediction. Furthermore, results showed that climate, land cover, and elevation variables are more important than anthropogenic variables. The results of the effective variables are consistent with other studies by Pourtaghi et al. (2015) and Wu et al. (2014). Although rivers can mitigate fire severity directly and indirectly, a larger human community near rivers increases the likelihood of fire ignition. Thus, it can be considered as an intermediary anthropogenic factor. Picnic and camping areas are usually located near rivers, and we can categorize all of them as recreational spaces. Therefore, the distance to rivers or recreational areas is often treated as a single factor in many studies (Tehrany et al. 2021; Pourghasemi et al. 2020; Moayedi et al. 2020; Jaafari et al. 2019b). Kalantar et al. (2020) highlighted the importance of the distance to rivers in wildfires, ranking as the second most important factor among 14 variables. Bloch et al. (2013a, b) indicated a high likelihood of ignition from campfires and recreational activities along the Methow and Chewuch rivers and the Pearrygin Lake State Park in the Okanogan region. Other studies considered only the main direct factors, such as distance to roads or housing, in their models and ignored factors like rivers, which can have secondary effects (Fusco et al. 2016; Faivre et al. 2014). Thus, the river's indirect role in wildfires depends on the physical characteristics of the region. Results of this work show that the distance to rivers has a very low impact in both study areas, and we can consider them as a secondary and indirect factor. In addition, findings revealed that about 45% of the historical wildfires in the Okanagan region occurred at an elevation exceeding 1542 m, and over 90% and 99% of fire incidents were localized at a considerable distance from roadways in the Okanagan and Jamésie regions, respectively. Thus, results can highlight that anthropogenic variables (e.g., through access to the forest) exert a negligible impact on the occurrence of wildfires in these regions. In other words, anthropogenic factors play an indirect role in wildfire occurrence compared to natural factors in the case studies. Climate change, caused by human-induced greenhouse gas emissions, exacerbates conditions favorable for wildfires, including prolonged droughts and higher temperatures. Human activities, such as land development, agriculture, and forestry strategies, often disrupt natural ecosystems, making them more susceptible to fire. Additionally, increased ignition sources from human negligence, such as campfires left unattended, contribute to wildfire occurrence. The anthropogenic and lightning ignitions, as two main sources of wildfires, affect fire regimes differently. Humans mainly impact fire regimes by changing the distribution and density of ignitions, the seasonality of burning, and available fuels (Balch et al. 2017). In the United States, human ignitions have significantly lengthened the wildfire season by more than three times that caused by lightning, especially in the spring. Furthermore, human-caused fires are the primary contributors to large wildfires in the Western U.S. (Barros et al. 2021). Unlike the Western U.S., large human-caused fires seem to be declining in Canada (Hanes et al. 2018). Nevertheless, human ignitions remain a crucial issue, accounting for about half of all large wildfires and typically occurring closer to populated areas (Johnston and Flannigan 2018). There are also distinct seasonal patterns between lightning- and human-ignited wildfires (Coogan et al. 2020). In Canada, lightning-caused fires are more prevalent during the summer months (Burrows and Kochtubajda 2010), while human-caused fires can increase during the summer and generally peak in spring months like May (Wotton et al. 2010).

To examine the temporal and spatial generalization ability of the developed models, datasets from different times and locations are used for training and validation of the LR and RF. Results showed that the developed models trained in a specific time and location could be used to predict wildfires in another time and location. Although the same performance and suitability of the models is not expected using different training sets in the time and location, the models' temporal and spatial generalization capability is crucial, particularly for regions lacking (proper) wildfire data. Note that machine learning as a data-driven algorithm strongly depends on the data. Thus, more valid and related data can improve the performance and skill of the models.

The RF model generally outperforms the LR model in wildfire prediction (particularly in Jamésie), which is consistent with other studies (Gholamnia et al. 2020; Guo et al. 2016). Indeed, RF is a widely used, accurate, and efficient algorithm compared to other well-known ML algorithms (Jain et al. 2020). It is more interpretable with low computational cost that makes it suitable for handling high-dimensional datasets, and thus it is reasonable for predictive modeling. In addition, spatial generalization results indicate that although both models have the generalization ability to be trained in one region and used for another region, training sets from the Okanogan region can lead to a better result, probably due to more variability in the training data. For instance, the average elevation in the Jamésie region is 208 m, with an absolute minimum elevation of zero meters, while there is a markable variation in elevation, ranging from 181 to 2712 m in the Okanogan region. This variability can strongly affect temperature and precipitation. Thus, using more data with higher variability in training the models can lead to a more robust model.

Our analysis showed that both study areas experienced a prevalent occurrence of wildfires in locales with lower mean temperatures. Although high temperatures can trigger many wildfires, relatively lower mean temperatures lead to wildfires in these study areas. In addition, findings revealed that in the Okanogan region, most wildfires occurred in the regions with decreased average precipitation rates, which provides dry conditions and thus increases susceptibility to fire. This is consistent with the results by Holden et al. (2018), which showed that a decrease in summer precipitation contributed to the increase in wildfires over the past three decades. They showed that in some parts of the Western United States, decreasing summer precipitation and fewer rain days can be the primary drivers of increased wildfire activity due to enhancing sensible heat and vapor pressure deficit that can lead to higher fuel aridity. In addition, Stephanie et al. (2020) indicated that increasing temperature and vapor pressure deficit with declining precipitation are linked to an increase in the area burned by wildfires across the Western United States. However, many studies focused on just temperature and linked recent trends in fire activity in the Western United States to rising temperatures, with warmer winter temperatures that can lead to reduced snow accumulation and an extended fire season (Westerling, 2003). Additionally, higher summer temperatures increase evaporation, which dries out woody fuels (Williams et al. 2013; Abatzoglou & Williams 2016). On the other hand, there is another mechanism for the occurrence of wildfires in Jamésie. Most wildfires occur in areas with significantly higher average rainfall rates, which can accelerate vegetation growth as a potential fuel for future wildfires (Li et al. 2023). Results confirm that different factors and variables, depending on the locations, can trigger and increase wildfires. The interaction of various factors with different levels of importance makes the models complex, which highlights the importance of datasets, training, and physical analysis of the system. The probabilities of the wildfire occurrence as the outputs of the models are used to provide susceptibility fire maps with the severity levels, including very low (average from 0 to 0.16), low (average from 0.16 to 0.37), medium (average from 0.37 to 0.59), high (average from 0.59 to 0.78), and very high (average from 0.78 to 1). Results indicated that higher elevated areas of the Okanogan and Jamésie regions have higher severity levels and vulnerability to fire. These findings highlight critical domains that are substantial for mitigating fire hazards and reducing damage and losses.

The development of wildfire prediction models and fire susceptibility maps provides valuable sources for managing and mitigating increased wildfire occurrences globally. The model can be developed to assess fire spread and growth behavior as the future area of research since modeling the spread of wildfires is needed for fire regulation agencies, for instance, for proper allocation of suppression resources and evacuation planning. ML algorithms significantly improve the accuracy and efficiency of wildfire hazard assessment, which can be used for risk analysis and mitigation strategies. Ecologists can identify areas at higher risk to protect vulnerable areas and conserve biodiversity. Results and analysis of this work can be applied in other regions to find vulnerable areas to conserve ecosystems and protect infrastructures and communities.

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Authors' contributions
Sanaz Moghim: supervision; conceptualization; investigation; methodology; analysis; validation; visualization; writing. Majid Mehrabi: methodology; software; investigation; analysis; validation; data curation; visualization; writing.

Data availability
Data are publicly available; related links are added in the manuscript.

Declarations

Competing interests
The authors declare no competing interests.

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